

Ross Hillman

hillmanross@gmail.com | (919) 667-4366 | [linkedin.com/in/ross-hillman/](https://www.linkedin.com/in/ross-hillman/)

SKILLS

Software/Technologies: Excel, FlexSim, Power BI, CAD, JMP, Visio, Power Query, Simio, Tableau, BlueBeam

Programming Languages: SQL, VBA, C#, Python, C++

Competencies: Data Analysis, Data Modeling, Discrete Event Simulation, Forecasting, Capacity Planning, Lean, Six Sigma, Project Management, Traffic Analysis, Process Improvement, Statistical Process Control, Entrepreneurial Thinking, Root Cause Analysis, Continuous Improvement

WORK EXPERIENCE

Industrial Engineer at CRB Group | *Life Science Consulting* | July 2025 - Present

- Quantified benefits of automation tools in conceptual and basis of design phases for facility construction.
- Iteratively validated high-level manufacturing process designs from senior fellows using discrete event simulation to inform feasibility and weighed design choices through sensitivity analysis.
- Collaborated with process and automation engineers to design state-of-the-art production & storage operations, and inform architectural design of greenfield manufacturing facility.
- Verified suspected bottleneck within current operations through simulation, then ran multiple scenarios exploring different combinations of solutions to relieve current bottleneck and provided cost-benefit analysis of solutions.

Industrial Engineer at Wolfspeed | *Semiconductor Manufacturing* | October 2023 - November 2024

- Investigated tool & material usage statistics using MES and CMMS log data to program statistical models of factory capacity to forecast KPIs, updating as necessary to capture technical process changes and continuous improvement projects.
- Presented monthly capacity reports to leadership covering equipment availability, utilization, bottlenecks, sales & operations PSI plan implications, and relevant projects effects & risks.
- Led working sessions with process, product, and equipment engineers to document institutional knowledge, quantify cost/benefit of technical changes, and guide project plan implementation.
- Modeled and presented several scenarios to leadership on reducing costs to meet cash flow targets, resulting in delaying \$2.2 million dollars in CAPEX to a preferable time, and executing projects saving \$600k/year in OPEX.

Industrial Engineer at CRB Group | *Life Science Consulting* | May 2022 - September 2023

- Analyzed historical data combined with future demand growth projections to right-size dock, warehouse, and cold room storage space.
- Performed high-level observation & interview-based study to locate areas of improvement within current process, then made recommendations on future studies to analyze root causes and explore possible solutions.
- Utilized expansion designs including floorplans, layouts, and schedules to construct a traffic simulation to inform feasibility of design plans, optimal routes within specified constraints, and areas prone to congestion.
- Created a capacity model to determine the practicality of a central weigh & dispense station based on anticipated demand, quantified benefits/time savings, then ranked possible locations based on adjacencies, cost and disruption based on current operations.
- Leveraged equipment dedications, historical production data and future projections to run discrete-event simulations. Performed sensitivity analysis on output data to right-size equipment before capital expenditure.

EDUCATION

Industrial Engineering B.S. | North Carolina State University

- Business Administration Minor
- Study abroad at Seoul National University in South Korea